PHISHING EMAIL
DETECTION & AWARENESS REPORT

Cyber Security Internship - Task 2
Future Interns

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Organization	Future Interns
Report Date	20 April 2026
Classification	Confidential
Scope	Phishing Email Analysis & Awareness

Official Internship Report

This report is prepared for educational purposes as part of a Cyber Security Internship.

1. Executive Summary

This report presents a detailed analysis of four phishing email samples collected and examined as part of a Cyber Security Internship. The objective of this report is to identify common phishing indicators, classify email risk levels, and provide practical awareness guidelines to help individuals and organizations protect themselves from phishing attacks.

Phishing is one of the most common and costly cyber threats faced by businesses today. Attackers trick users into revealing sensitive information, clicking malicious links, or transferring funds by impersonating trusted entities via email.

Emails Analyzed	Phishing Confirmed	Attack Types	Risk Level
4	**4/4**	**4 Types**	**High**

2. Scope & Methodology

2.1 Objective

- Analyze phishing email samples for common indicators
- Classify each email by risk level: Safe / Suspicious / Phishing
- Explain how each attack works in simple, non-technical language
- Provide practical prevention tips for users and employees

2.2 Tools Used

Tools	Purpose
Manual Email Analysis	Identifying phishing indicators in email content
Browser DevTools	Inspecting links and domains safely
HTMLEmail.io	Rendering email samples for evidence

2.3 Ethical Guidelines

All phishing email samples analyzed in this report are simulated examples created strictly for educational and research purposes as part of a Cyber Security Internship. No real phishing emails were sent, no systems were attacked, and no individuals were harmed. All analysis was conducted passively and ethically.

3. Common Phishing Indicators

The following indicators were used to identify and classify phishing emails throughout this report:

Sr.	Indicator	Risk	Description
1.	**Fake / Spoofed Sender Domain**	**High**	Email appears from a trusted brand but domain is slightly altered (e.g. paypa1.com instead of paypal.com)
2.	**Urgency & Fear Language**	**High**	Uses phrases like 'Act immediately', 'Account will be locked', or '24 hours' to pressure victims
3.	**Generic Greeting**	**Medium**	Uses 'Dear User' or 'Dear Customer' instead of the recipient's real name
4.	**Suspicious / Malicious Links**	**High**	Links appear legitimate but redirect to fake websites designed to steal credentials
5.	**Credential Harvesting**	**Critical**	Asks the user to enter their current password, OTP, or banking details
6.	**Secrecy Demand**	**High**	Instructs victim not to share the email or discuss with colleagues to prevent verification
7.	**Too Good To Be True**	**High**	Offers unexpected rewards like loan approvals, prizes, or job offers to lure victims
8.	**Blocks Verification**	**High**	Asks victim not to call or verify through official channels to avoid detection
9.	**Advance Fee Fraud**	**Critical**	Asks for a small upfront payment (e.g. processing fee) to release a larger reward
10.	**Business Email Compromise**	**Critical**	Impersonates a CEO or executive to authorize fraudulent wire transfers

4. Sample 1 - Fake PayPal Security Alert

Email Type	Brand Impersonation / Credential Phishing
Sender Domain	service@paypa1-secure-login.com
Subject Line	Urgent: Your PayPal account will be locked because suspicious activity detected on your account!
Risk Classification	Phishing - Critical Risk

Red Flags Identified

Sr.No	Red Flag	Severity	Explanation
1.	**Fake Sender Domain**	**High**	Domain uses '1' instead of 'l' in PayPal - a classic typosquatting technique
2.	**Urgency / Fear Language**	**High**	Subject line uses **Urgent** and **threatens** account lockout within 24 hours
3.	**Generic Greeting**	**Medium**	'Dear User' used instead of the recipient's actual name
4.	**Malicious Link**	**High**	Link points to paypal-secure.verify-me.com - real domain is verify-me.com, not PayPal
5.	**Aggressive Tone**	**Medium**	'Act immediately!' and 'permanent termination' used to panic the victim

How This Attacks Works

The attacker impersonates PayPal by slightly modifying the domain name (paypa1 instead of paypal). The email creates panic by threatening account suspension, then directs the victim to a fake website that looks identical to PayPal's login page. When the victim enters their credentials, the attacker captures them and gains full access to the victim's real PayPal account.

Business Impact

- Unauthorized access to PayPal account and linked bank accounts
- Financial theft through unauthorized transactions
- Personal and financial data exposure
- Identity theft using stolen credentials

Prevention Tips

- Always check the sender's email domain carefully before clicking any links
- Go directly to PayPal.com by typing it in the browser - never use email links
- Enable two-factor authentication (2FA) on all financial accounts
- PayPal will never ask you to verify via an unsolicited email link

Verdict: Phishing - Critical Risk

PayPal

Security Notification Center

From: service@paypa1-secure-login.com
To: user@gmail.com
Subject: Urgent: Your PayPal account will be locked because suspicious activity detected on Your Account!
Date: Sun, 13 Apr 2026 09:41:00 GMT

Dear User,

We have detected suspicious login activity on your **PayPal** account from an unrecognized device. To protect your account, access has been temporarily limited.

Your account will be permanently suspended **within 24 hours** unless you verify your identity immediately.

Please click the button & link below to verify your account details and restore full access:
http://paypal-secure.verify-me.com/account/confirm?user=you&token=8f3kd92

Verify My Account Now

Failure to verify will result in permanent account termination. Act immediately!

If you believe you received this in error, you may disregard this message.

Regards,
PayPal Security Team

© 2026 PayPal Inc. | 2211 North First Street, San Jose, CA 95131
This email was sent to user@gmail.com

5. Sample 2 - Fake IT Department Password Reset

Email Type	Internal Phishing / Credential Harvesting
Sender Domain	it-support@yourcompany-helpdesk.net
Subject Line	Reset Your Password Now
Risk Classification	Phishing - Critical Risk

Red Flags Identified

Sr.No	Red Flag	Severity	Explanation
1.	**Fake Internal Domain**	**High**	Domain yourcompany-helpdesk.net mimics an internal IT address but is externally registered
2.	**Urgency Tactic**	**High**	'Password expires in 2 hours' creates artificial time pressure to bypass rational thinking
3.	**Generic Greeting**	**Medium**	'Dear Employee' used - real IT departments address employees by name
4.	**Credential Harvesting**	**Critical**	Asks employee to enter their CURRENT password - IT teams never need your current password
5.	**Malicious Reset Link**	**High**	Link points to an external domain, not the company's official IT portal

How This Attacks Works

The attacker registers a domain that closely resembles the company's IT helpdesk. They send an email pretending to be the internal IT team, creating urgency about password expiry. The victim clicks the link and lands on a fake company login page. When they enter their current and new passwords, the attacker captures both - gaining full access to company systems, email, and internal tools.

Business Impact

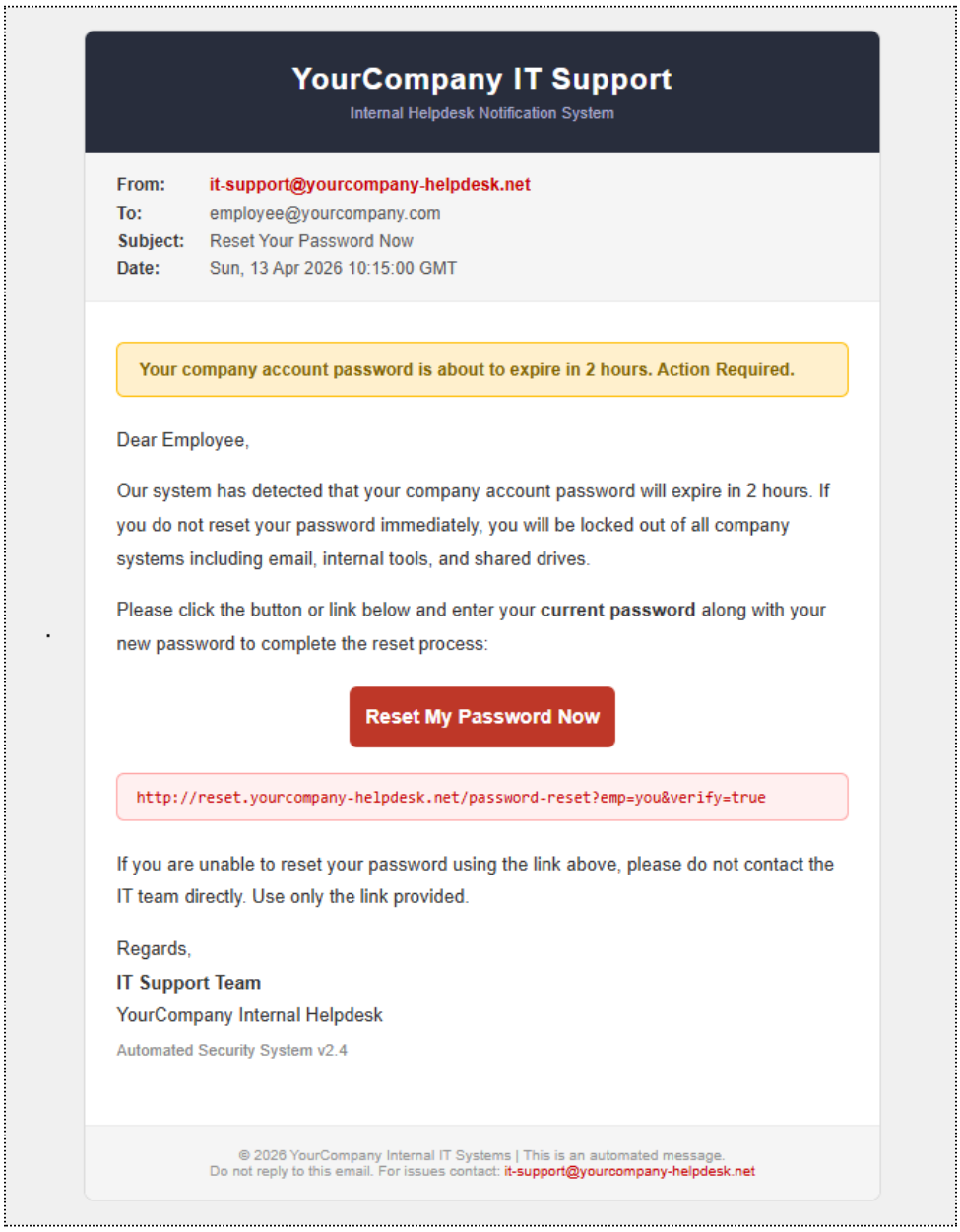
- Complete compromise of employee's corporate account
- Unauthorized access to company email, files, and internal systems

- Potential data breach affecting entire organization
- Lateral movement - attacker can target other employees using compromised account

Prevention Tips

- Legitimate IT departments NEVER ask for your current password
- Always verify password reset requests by calling IT directly using the official number
- Access password reset only through the official company intranet portal
- Report suspicious internal emails to your security team immediately

Verdict: Phishing - Critical Risk



YourCompany IT Support

Internal Helpdesk Notification System

From: it-support@yourcompany-helpdesk.net
To: employee@yourcompany.com
Subject: Reset Your Password Now
Date: Sun, 13 Apr 2026 10:15:00 GMT

Your company account password is about to expire in 2 hours. Action Required.

Dear Employee,

Our system has detected that your company account password will expire in 2 hours. If you do not reset your password immediately, you will be locked out of all company systems including email, internal tools, and shared drives.

Please click the button or link below and enter your **current password** along with your new password to complete the reset process:

Reset My Password Now

http://reset.yourcompany-helpdesk.net/password-reset?emp=you&verify=true

If you are unable to reset your password using the link above, please do not contact the IT team directly. Use only the link provided.

Regards,
IT Support Team
YourCompany Internal Helpdesk
Automated Security System v2.4

© 2026 YourCompany Internal IT Systems | This is an automated message.
Do not reply to this email. For issues contact: it-support@yourcompany-helpdesk.net

6. Sample 3 - Fake Bank Loan Approval

Email Type	Advance Fee Fraud / Financial Phishing
Sender Domain	loans@nationaltrustbank-verify.com
Subject Line	Congratulations! Your Loan of $25,000 Has Been Approved
Risk Classification	Phishing - Critical Risk

Red Flags Identified

Sr.No	Red Flag	Severity	Explanation
1.	**Fake Bank Domain**	High	nationaltrustbank-verify.com is not a real bank domain - 'verify' is appended to seem legitimate
2.	**Unsolicited Offer**	High	Victim never applied for a loan - unexpected approvals are a classic social engineering tactic
3.	**Generic Greeting**	Medium	'Dear Valued Customer' instead of the recipient's real name
4.	**Advance Fee Fraud**	Critical	Asks for a $50 'processing fee' to release $25,000 - victim pays and receives nothing
5.	**Urgency Tactic**	High	'Offer expires in 24 hours' pressures the victim to act without thinking
6.	**Malicious Claim Link**	High	Button links to an external phishing site designed to steal banking details
7.	**Secrecy Demand**	High	'Keep this email confidential' prevents victim from seeking advice

How This Attacks Works

This attack uses the lure of unexpected financial gain to trick victims. The attacker impersonates a bank and sends a fake loan approval for a large amount ($25,000). However, to 'release' the funds, the victim must pay a small processing fee ($50) or provide their bank account details. Once the fee is paid or details are shared, the attacker disappears with the money and the victim receives nothing in return.

Business Impact

- Direct financial loss from paying fake processing fees
- Bank account compromise if account details are provided
- Identity theft using provided personal and financial information
- Psychological distress and loss of trust in financial institutions

Prevention Tips

- Never pay upfront fees to receive money - legitimate loans do not work this way
- Verify any loan approval directly with the bank using their official website or phone number
- Be suspicious of any unsolicited financial offers, no matter how convincing
- Check if the bank domain matches the official bank website exactly

Verdict: Phishing - Critical Risk

National Trust Bank

Customer Loan Services Department

From: loans@nationaltrustbank-verify.com
To: customer@gmail.com
Subject: Congratulations! Your Loan of $25,000 Has Been Approved
Date: Sun, 13 Apr 2026 11:30:00 GMT

Congratulations! Your personal loan application has been approved!

Dear Valued Customer,

We are pleased to inform you that your loan application has been successfully approved by our financial team. Please review your loan details below:

Loan Amount	$25,000 USD
Loan Type	Personal Loan
Interest Rate	3.5% per annum
Repayment Period	36 months
Reference No	NTB-2026-APR-88432
Approval Date	13 April 2026

To claim your approved loan amount, you must verify your bank account details and pay a small processing fee of $50 within 24 hours to release the funds:

Click the link here to claim approved loan.

Claim My Loan Now

Please keep this email confidential and do not share your reference number with anyone.

Regards,
Mr. Robert Collins
Senior Loan Officer
National Trust Bank - Customer Services
Tel: +1-800-555-0192 | loans@nationaltrustbank-verify.com

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7. Sample 4 - Fake CEO Wire Transfer (BEC Attack)

Email Type	Business Email Compromise (BEC) / CEO Fraud
Sender Domain	james.mitchell@techcorp-intl.net
Subject Line	Urgent: Confidential Wire Transfer Required Today
Risk Classification	Phishing - Critical

Red Flags Identified

Sr.No	Red Flag	Severity	Explanation
1.	**CEO Domain Spoofing**	**High**	techcorp-intl.net instead of techcorp.com - subtle difference most employees miss
2.	**Extreme Urgency**	**High**	'Must be completed before 3PM or deal falls through' - no time to verify
3.	**Secrecy Demand**	**High**	'Do not discuss with anyone' isolates the victim and prevents verification
4.	**Blocks Phone Verification**	**High**	'Do not call me' - prevents the employee from confirming with the real CEO
5.	**Large Wire Transfer Request**	**Critical**	Requests $87,500 - attackers target finance staff who have transfer authority
6.	**Social Engineering**	**High**	Uses real-sounding names, titles, and meeting context to appear completely legitimate
7.	**No Malicious Links**	**High**	Relies purely on social engineering - no links or attachments to trigger security tools

How This Attacks Works

BEC (Business Email Compromise) is the most sophisticated and costly form of phishing. The attacker researches the company, identifies the CEO and a finance employee, then sends a spoofed email from an address nearly identical to the CEO's. There are no malicious links or attachments - the email looks completely legitimate. The finance employee, believing it is from

the CEO, processes the wire transfer. By the time the fraud is discovered, the money has been moved through multiple accounts and is unrecoverable.

Business Impact

- Direct financial loss - average BEC attack costs $125,000 per incident globally
- Funds are often unrecoverable once transferred internationally
- Reputational damage to the organization
- Legal and regulatory consequences for the company
- Employee psychological impact - the victim often faces disciplinary action

Prevention Tips

- Always verify large wire transfer requests by calling the CEO directly using an official number
- Implement a two-person approval policy for all wire transfers above a threshold
- Train all finance staff to recognize BEC attacks specifically
- Never process financial requests based solely on email - always verify through a second channel
- Implement DMARC, DKIM, and SPF email authentication to prevent domain spoofing

Verdict: Phishing - Critical Risk

JM

James Mitchell

Chief Executive Officer - TechCorp International

High Priority - Confidential - Immediate Action Required

From: james.mitchell@techcorp-intl.net
To: finance@techcorp.com
Subject: Urgent: Confidential Wire Transfer Required Today
Date: Sun, 13 Apr 2026 08:05:00 GMT

Hi Sarah,

I am currently in a confidential board meeting and unable to take calls. I need you to process an urgent wire transfer immediately for a time-sensitive acquisition deal closing today.

Please process the following transfer details as soon as possible:

Transfer Amount:	**$87,500 USD**
Beneficiary Name:	Global Acquisitions LLC
Bank Name:	First International Bank
Account Number:	4829-3847-2910-0038
Routing Number:	021000021
Reference:	TCORP-ACQ-2026-APR

Do not discuss this transfer with anyone in the office. This is a confidential acquisition and must remain private until formally announced.

Please confirm via email once the transfer is complete. Do not call me - I am in meetings all day and cannot be disturbed.

8. Phishing Awareness Guidelines

8.1 The Golden Rules - For All Employees

- STOP - Before clicking any link, take a moment to examine the email carefully
- CHECK - Verify the sender's email domain matches the official company domain exactly
- THINK - Ask yourself: Did I request this? Is this expected? Does it seem too urgent?
- VERIFY - When in doubt, contact the sender through a separate, official channel
- REPORT - Report any suspicious emails to your IT or security team immediately

8.2 How to Identify the Phishing Email

Warning Signs	What to Do
Sender domain looks slightly wrong	Check every letter - paypa1.com is NOT paypal.com
Email asks for your password	Legitimate services NEVER ask for your password via email
Urgent threats or deadlines	Slow down - urgency is designed to make you skip verification
Unexpected financial offers	If you didn't apply for it, you didn't win it - it's a scam
Links that look suspicious	Hover over links first - check if the real URL matches
'Don't tell anyone' instructions	Legitimate requests are never secret - this is a red flag

8.3 Do’s & Don’ts for Employees

Do	Don’t
Verify sender domain carefully	Don't click links in unexpected emails
Enable 2FA on all accounts	Don't enter passwords on links from emails
Call to verify unusual requests	Don't wire money based only on an email
Report phishing to IT team	Don't download unexpected attachments
Type URLs directly in browser	Don't share OTPs or passwords via email

9. Conclusion

This report analyzed four distinct phishing email samples representing common attack types encountered in real-world scenarios. All four emails were confirmed as phishing attempts exhibiting multiple red flags including domain spoofing, urgency tactics, credential harvesting, advance fee fraud, and business email compromise.

The analysis demonstrates that phishing attacks range from simple brand impersonation to highly sophisticated social engineering techniques like BEC attacks that contain no malicious links at all. This highlights the critical importance of user awareness and verification procedures as the primary line of defense against phishing.

Organizations and individuals can significantly reduce their risk by implementing the awareness guidelines outlined in this report, adopting multi-channel verification procedures for sensitive requests, and maintaining a culture of security awareness.

Sample	Attack Type	Risk	Verdict
Sample 1	Brand Impersonation (PayPal)	Critical	Phishing
Sample 2	IT Credential Harvesting	Critical	Phishing
Sample 3	Advance Fee Fraud (Bank)	Critical	Phishing
Sample 4	Business Email Compromise	Critical	Phishing

End of Report

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